Parallel Programming: Assignment 2

Delta-Stepping Single Source Shortest Path on GPU

Implement parallel delta-stepping algorithm for finding Single Source Shortest Path, dealt in the class, on GPU with CUDA. Implement GPU optimizations that are applicable, e.g., coalesced memory access. Check the correctness of the results with a sequential CPU implementation.

Show results for the graph given in the below link: `https://drive.google.com/file/d/17mzIlsWRUclzwOzQlwgP9view?usp=sharing`

Implement on the course's GPU node. Perform experiments with source vertices as 0, 500, 1000, 10000, 50000 and 100000. Show comparisons of execution times with the CUDA version vs the sequential version. Also, for CUDA execution, run with CUDA profiler, show and analyze the profiling outputs. For details on the CUDA profiler, see `https://docs.nvidia.com/cuda/profiler-users-guide/index.html`